

MANOJ ADHIKARI

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Education

Ph.D., Computer Science, Kennesaw State University Aug 2026 - Present |GA, USA
Research Area: AI for Educational Technology/Learning Science.
M.S., Computer Science, East Tennessee State University, GPA:4.0/4.0 2023 - 2025 |TN, USA
B.S., Computer Science, Tribhuvan University, Percentage: 80.12% (Distinction) 2016 - 2020 |KTM, Nepal

Publications (C = Conference)

- [C1] **M. Adhikari**, P. Joshi, G. V. Ramos, A. A. Doulat and S. Shaik, "AIDE: Leveraging Retrieval-Augmented Generation for Context-Aware Educational Data Retrieval and Dialogue," 2025 International Conference on Smart Applications, Communications and Networking (SmartNets), Istanbul, Turkiye, 2025, pp. 1-7, doi: 10.1109/SmartNets65254.2025.11106900. [view](#)
- [C2] **M. Adhikari**, M. Prajapati, P. Adhikari, and S. Shaik, "PALM: Phoneme-Level Prosody and Lexical Feature Augmented Dataset Construction Pipeline for Language and Speech Studies," 2026 International Conference on Smart Applications, Communications and Networking (SmartNets), Rome, Italy, 2026, pp. 1-8, doi: 10.1109/SmartNets69662.2026.11604988. [view](#)
- [C3] **M. Adhikari**, P. Joshi, S. Shrestha and S. Shaik, "Vision-Based Driver Drowsiness and Distraction Detection through Behavioral Indicators of Fatigue," SoutheastCon 2025, Concord, NC, USA, 2025, pp. 261-266, doi: 10.1109/SoutheastCon56624.2025.10971515 [view](#)
- [C4] P. Joshi, **M. Adhikari**, S. Shrestha and S. Shaik, "Real-Time Driver Drowsiness Detection Using CNN, MediaPipe, and ML Classifiers," SoutheastCon 2025, Concord, NC, USA, 2025, pp. 589- 594, doi: 10.1109/SoutheastCon56624.2025.10971270 [view](#)
- [C5] C. Young, A. A Doulat, P. Joshi, **M. Adhikari**, and S. Kshatri, "Modeling Student's Emotional Trajectories in Course Evaluations: A Computational Framework for Enhancing Student Success," OTESSA'25, In Press.

Research Experience

AI in Educational Technology (AIET) Lab, Kennesaw State University Aug 2026 – Present
Project: ProgCT-Computational Thinking in Programming Advisor: Dr. Nasrin Dehbogorzi

- Researched the integration of computational thinking (CT) into programming instruction within AI-supported learning systems, emphasizing decomposition, abstraction, pattern recognition, and algorithm design.
- Developed structure-aware code representations using AST parsing (Tree-Sitter), Graph Neural Networks (GCN), and semantic embeddings to support analysis of programming solutions through a CT-focused lens.
- Built a hybrid retrieval pipeline combining dense embeddings, BM25 lexical scoring, and FAISS indexing to retrieve programming examples relevant to instructional concepts and learner queries.
- Implemented query-adaptive weighting and re-ranking to connect conceptual learning questions with relevant code structures and algorithmic patterns.
- Deployed quantized local LLMs and evaluated retrieval-generation performance to support the development of CT-oriented explanations and instructional feedback in programming learning systems.

Computing Lab, East Tennessee State University Sep 2024 - Jun 2026
Project: Hybrid Code Search Framework Advisor: Dr. Shehenaz Shaik

- Designed a structure-aware code retrieval engine integrating AST parsing (Tree-Sitter), Graph Neural Networks (GCN), MiniLM embeddings, BM25 lexical scoring, and FAISS indexing.
- Built a hybrid retrieval pipeline combining graph-based code representations with dense semantic and sparse lexical signals for context-aware discovery of functions and algorithms.
- Compiled and quantized local LLM models for efficient deployment, and evaluated retrieval-generation performance under constrained inference settings.

- Implemented dynamic query-adaptive weighting and re-ranking to improve precision across concept-level and syntax-specific developer queries.
- Developed an end-to-end workflow supporting LLM-assisted software engineering tasks including code understanding, reuse, and bug localization.

- Study and design information-retrieval architectures that support robust, context-aware decision-making.
- Developed and evaluated prompt chains and query-rewriting templates, to ensure content safety and preserve contextual fidelity across multi-turn dialogue.

Fr. Lockie & Stiller Research Program, St. Xavier's College 2018 - 2019
Crime Risk Assessment - Kathmandu Valley Advisor: Lecturer Rajan Karmacharya

Grants, Awards & Honors (G = Grant, A = Award, H = Honor)

Fellowship & Volunteering

Work Experience

- Instructed and led lab sessions in person classes for CSCI 1100/1150 (Using Information Technology).
- Collaborated with the professor to develop course materials on topics of Data, AI, and Cybersecurity literacy.
- Graded assignments and held regular office hours to provide academic support and guidance to students.

- Developed LLM-powered agents to automate insurance workflows, integrating tool calling, contextual retrieval, and multi-step task execution.
- Built and maintained Model Context Protocol (MCP) servers connecting AI agents to internal APIs, databases, and business systems.
- Optimized MCP tool schemas, request handling, and context delivery to improve tool selection, reduce latency, and manage token consumption.
- Implemented agent orchestration, state management, retries, and fallback mechanisms to support reliable execution of complex workflows.
- Established evaluation and monitoring workflows to assess response quality, tool execution accuracy, latency, and inference costs.

Skills

Technical: Python, C, C++, Java, SQL, NoSQL, Vector Databases, Git, Docker, Apache Airflow, TensorFlow, Keras, PyTorch, LangChain, LLMs, Transformers, RAG Systems, Knowledge Graphs, Neural IR, Embeddings, Transfer Learning, Multimodal Learning, Data Preprocessing, Model Optimization, Evaluation Benchmarking .
Research: Experimental Design, Statistical Modeling, Quantitative Analysis, Qualitative Methods, Ablation Studies, Error Analysis, User Study Design, Scientific Writing, Reproducible Research.

Professional Organization

IEEE Computer Society, Young Professional, Intelligent Informatics Member

References (R = Relationship)

Dr. Nasrin Dehbozorgi
Assistant Professor, Kennesaw State University dnasrin@kennesaw.edu [R]: Ph.D Advisor

Dr. Abhishek Parakh
Professor, Kennesaw State University, aparakh@kennesaw.edu [R]:Advisor

Dr. Shehenaz Shaik
Assistant Professor, East Tennessee State University, shaiks@etsu.edu [R]: Research Supervisor

Dr. Chelsea Dubay
Assistant Professor, East Tennessee State University, dubayc@etsu.edu [R]: TA Supervisor

Human Subjects Research Certifications

- Students in Research, Citi Link
- Socio-Behavioral-Educational Researchers, Citi Link

Selected Projects GitHub

Interakt

Tools: Python, Streamlit, YOLOv8, MediaPipe

- Designed and implemented a real-time daily activity tracking and behavioral analytics system integrating YOLOv8 for object detection and MediaPipe for human-pose estimation.
- Developed a continuous webcam-based monitoring pipeline to detect user interactions with everyday objects, generating structured, time-stamped activity logs for downstream analysis.
- Built data-parsing modules to extract temporal patterns, interaction frequencies, and contextual cues from activity sequences.
- Conducted statistical and exploratory analyses to derive behavioral insights such as distraction indicators, multi-tasking patterns, and intentional vs. unintentional user engagement.
- Evaluated system performance under varying lighting and motion conditions and optimized object detection thresholds to improve reliability and reduce false positives.

Project: MetaMorph-Deepfake — Metamorphic Testing Framework for Deepfake Detection Tools: Python, PyTorch, OpenCV, Albumentations

- Designed a metamorphic testing framework to evaluate robustness of deepfake detection models under semantic-preserving transformations (compression, noise, blur, rotation, and color perturbations).
- Implemented automated adversarial data pipelines to generate transformed video and image samples for stress-testing model invariance and generalization.
- Conducted systematic evaluation across multiple CNN-based detectors to quantify performance degradation and uncover hidden failure modes under distribution shifts.
- Developed reproducible benchmarking workflows to analyze robustness gaps, supporting trustworthy AI assessment and model validation.

- Provided actionable insights for improving detector resilience by identifying transformation-sensitive features and vulnerability patterns.

Phishing Awareness and Simulation System

Tools: Python, Seaborn, Scikit-learn

- Architected backend REST APIs for an intelligent phishing-simulation platform designed to educate users and measure organizational cybersecurity awareness.
- Developed realistic phishing templates with variable difficulty levels and built automated campaign workflows that track browser metadata, open rates, click patterns, and response behavior.
- Designed analytics modules to infer susceptibility profiles by analyzing user cohorts, behavioral triggers, and risk factors based on response patterns.
- Implemented one-pixel tracking for email engagement monitoring and integrated dark-web breach lookup endpoints to provide personalized risk assessments.
- Conducted statistical analysis to identify high-risk user groups and generate targeted training recommendations.

Project: S3 – Sensing Stress in Speech

Tools: Python, Librosa, PyTorch, Scikit-learn

- Developed an end-to-end speech stress detection pipeline extracting acoustic features (MFCCs, pitch, energy) from audio signals for real-time affective state sensing.
- Trained and evaluated machine learning and deep learning models (e.g., CNN, LSTM) to classify stress vs. neutral speech under varying noise and speaking conditions.
- Implemented signal preprocessing, feature normalization, and data augmentation to improve model robustness and generalization across speakers and environments.
- Performed quantitative evaluation using accuracy, F1-score, and confusion analysis to benchmark performance and identify modality-specific error modes.